

Questions for class discussion

You have read the paper “Smartphone Ownership, Age of Smartphone Acquisition, and Health Outcomes in Early Adolescence” by Barzilay et al., *Pediatrics*, 2026.

Discuss the following questions in groups. Some are the same as you prepared at home, and some are new. Come to an agreement and write down your answers.

Part 1: Understanding the Research Question

1. What is the primary research question of the paper?
2. What are the main exposures (independent variables) studied?
3. What are the health outcomes (dependent variables)?

Part 2: The study design

1. Did the researchers design a study and collect data to answer a research question?

2. Why did the researchers choose 12 years as their age of study?

3. How would you have designed a study to investigate a causal relationship between owning a smartphone and sleep duration, if you could do whatever you wanted? What are the practical issues of your study design?

Part 3: Testing differences in groups

1. Look at Table 1, and in particular to the line “Sleep duration”, what is the null and alternative hypothesis that authors are testing?

2. The two-samples t -test used by the authors is (approximately) defined as

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_2^2}{n_2}}}$$

Can you identify in the table the numbers that are used to compute the test statistics?

3. What does the reported p-value say about the tested hypotheses?
4. Table 1 is a descriptive summary of the raw data. Why do you think the researchers did not just report these results, but continued on with a regression model?
5. In the text, page 3 left column, the authors write “to correct for the testing of 3 outcomes, we applied a P-value threshold of 0.017 (0.05/3)”. Why is it necessary to correct for doing multiple tests? Did the researchers only do three tests? Suggest a stricter threshold that they also could have chosen.
6. Look at Figure 2A and the blue lines. What are the dots, and what are the lines? Why is there a vertical dotted line at 1? Which outcome is most related to smartphone ownership?